

TSIA2 Math Practice Test 5

PART A — QUESTIONS

TSIA2-M5-001

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A student pays \$1,260 for a 6-credit course. What is the cost per credit?

A) \$180

B) \$200

C) \$210

D) \$220

TSIA2-M5-002

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

Solve for x: $7x - 12 = 3x + 24$

A) 6

B) 8

C) 9

D) 12

TSIA2-M5-003

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A rectangular courtyard is 18 meters long and 11 meters wide. A walkway is built along one long side and one short side of the courtyard. How many meters of walkway are needed?

A) 29 meters

B) 36 meters

C) 47 meters

D) 58 meters

TSIA2-M5-004

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

The table shows the number of survey responses collected by a student group over four days.

Monday: 34

Tuesday: 41

Wednesday: 37

Thursday: 48

What is the mean number of responses collected per day?

- A) 38
- B) 39
- C) 40
- D) 41

TSIA2-M5-005

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A meal plan provides 18 lunches for \$153. At this rate, what would be the cost of 25 lunches?

- A) \$197.50
- B) \$204.00
- C) \$212.50
- D) \$225.00

TSIA2-M5-006

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

Which expression is equivalent to $3(4y + 5) - 2(3y - 7)$?

- A) $6y + 1$
- B) $6y + 29$
- C) $18y + 1$
- D) $18y + 29$

TSIA2-M5-007

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A circular table has a circumference of 16π inches. What is the diameter of the table?

- A) 8 inches
- B) 16 inches
- C) 32 inches
- D) 64 inches

TSIA2-M5-008

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A bag contains 9 yellow beads, 6 purple beads, and 5 silver beads. One bead is selected at random. What is the probability that the bead is yellow or silver?

- A) $\frac{7}{20}$
- B) $\frac{11}{20}$
- C) $\frac{7}{10}$
- D) $\frac{3}{4}$

TSIA2-M5-009

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A store increases the price of a desk from \$160 to \$188. What is the percent increase?

- A) 14%
- B) 17.5%
- C) 20%
- D) 28%

TSIA2-M5-010

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

A tutoring center charges \$18 for registration and \$16 for each tutoring session. If a student pays \$114 in total, how many tutoring sessions did the student attend?

- A) 5
- B) 6
- C) 7
- D) 8

TSIA2-M5-011

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A triangular lot has side lengths of 13 yards, 14 yards, and 15 yards. Its perimeter will be fenced. The fence costs \$9 per yard. What is the total cost of the fence?

- A) \$126
- B) \$252
- C) \$378
- D) \$504

TSIA2-M5-012

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

The data set shows the numbers of minutes a student spent exercising on seven days:

20, 25, 25, 30, 35, 40, 70

Which statement best describes the effect of the value 70 on the data set?

- A) It increases the mean more than it increases the median.
- B) It increases the median more than it increases the mean.
- C) It makes the mean and median equal.
- D) It decreases the range of the data set.

TSIA2-M5-013

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A package weighs 4.8 pounds. The shipping company charges \$2.35 per pound, rounded to the nearest pound before calculating the charge. What is the shipping charge?

- A) \$9.40
- B) \$11.28
- C) \$11.75
- D) \$12.20

TSIA2-M5-014

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

Which ordered pair is a solution to the system of equations?

$$y = 2x - 1$$

$$y = -x + 8$$

- A) (2, 3)
- B) (3, 5)
- C) (4, 7)
- D) (5, 9)

TSIA2-M5-015

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A rectangular room is 15 feet long and 12 feet wide. A scale drawing of the room uses 1 inch to represent 3 feet. What are the dimensions of the room on the scale drawing?

- A) 3 inches by 2 inches
- B) 4 inches by 5 inches
- C) 5 inches by 4 inches
- D) 45 inches by 36 inches

TSIA2-M5-016

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A committee has 5 first-year students and 7 second-year students. If two students are selected at random without replacement, what is the probability that both students are second-year students?

- A) $\frac{7}{22}$
- B) $\frac{7}{24}$
- C) $\frac{49}{144}$
- D) $\frac{1}{2}$

TSIA2-M5-017

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A phone battery drops from 92% charged to 56% charged in 3 hours. If the battery charge decreases at the same average rate, how many percentage points does it drop per hour?

- A) 9 percentage points per hour
- B) 10 percentage points per hour
- C) 12 percentage points per hour
- D) 36 percentage points per hour

TSIA2-M5-018

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

Which expression is equivalent to $x^2 - 16$?

- A) $(x - 4)(x - 4)$
- B) $(x + 4)(x + 4)$
- C) $(x - 4)(x + 4)$
- D) $(x - 8)(x + 2)$

TSIA2-M5-019

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A point is located at $(-4, 6)$. It is moved 7 units to the right and 9 units down. What are the coordinates of the new point?

- A) $(-11, 15)$
- B) $(-11, -3)$
- C) $(3, 15)$
- D) $(3, -3)$

TSIA2-M5-020

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A small business tracks the number of orders it receives during five weekdays.

Monday: 16

Tuesday: 19

Wednesday: 23

Thursday: 23

Friday: 29

Which statement is true?

- A) The median is 19 orders.
- B) The mode is 23 orders.

- C) The range is 23 orders.
D) The mean is 23 orders.

PART B — ANSWER KEY + EXPLANATIONS

TSIA2-M5-001 — Correct: C — Correct Answer: \$210 — Explanation: Divide the total cost by the number of credits: $\$1,260 \div 6 = \210 per credit.

TSIA2-M5-002 — Correct: C — Correct Answer: 9 — Explanation: Subtract $3x$ from both sides to get $4x - 12 = 24$. Add 12 to get $4x = 36$, so $x = 9$.

TSIA2-M5-003 — Correct: A — Correct Answer: 29 meters — Explanation: The walkway covers one long side and one short side, so the total length is $18 + 11 = 29$ meters.

TSIA2-M5-004 — Correct: C — Correct Answer: 40 — Explanation: The total is $34 + 41 + 37 + 48 = 160$. Dividing by 4 gives a mean of 40.

TSIA2-M5-005 — Correct: C — Correct Answer: \$212.50 — Explanation: The cost per lunch is $\$153 \div 18 = \8.50 . For 25 lunches, the cost is $25 \times \$8.50 = \212.50 .

TSIA2-M5-006 — Correct: B — Correct Answer: $6y + 29$ — Explanation: Distribute and combine like terms: $12y + 15 - 6y + 14 = 6y + 29$.

TSIA2-M5-007 — Correct: B — Correct Answer: 16 inches — Explanation: The circumference of a circle is $C = \pi d$. If $C = 16\pi$, then $d = 16$ inches.

TSIA2-M5-008 — Correct: C — Correct Answer: $7/10$ — Explanation: There are $9 + 6 + 5 = 20$ beads total. Yellow or silver beads total $9 + 5 = 14$, so the probability is $14/20 = 7/10$.

TSIA2-M5-009 — Correct: B — Correct Answer: 17.5% — Explanation: The increase is $\$188 - \$160 = \$28$. The percent increase is $28/160 = 0.175$, or 17.5%.

TSIA2-M5-010 — Correct: B — Correct Answer: 6 — Explanation: Subtract the registration fee: $114 - 18 = 96$. Divide by the session cost: $96 \div 16 = 6$ sessions.

TSIA2-M5-011 — Correct: C — Correct Answer: \$378 — Explanation: The perimeter is $13 + 14 + 15 = 42$ yards. At \$9 per yard, the cost is $42 \times 9 = \$378$.

TSIA2-M5-012 — Correct: A — Correct Answer: It increases the mean more than it increases the median. — Explanation: The value 70 is much larger than the other values, so it pulls the mean upward. The median remains the middle value, 30.

TSIA2-M5-013 — Correct: C — Correct Answer: \$11.75 — Explanation: The weight 4.8 pounds rounds to 5 pounds. The charge is $5 \times \$2.35 = \11.75 .

TSIA2-M5-014 — Correct: B — Correct Answer: (3, 5) — Explanation: Substitute $x = 3$ into both equations: $y = 2(3) - 1 = 5$ and $y = -3 + 8 = 5$, so (3, 5) satisfies both equations.

TSIA2-M5-015 — Correct: C — Correct Answer: 5 inches by 4 inches — Explanation: Divide each actual dimension by 3: $15 \div 3 = 5$ inches and $12 \div 3 = 4$ inches.

TSIA2-M5-016 — Correct: A — Correct Answer: $7/22$ — Explanation: The probability of selecting a second-year student first is $7/12$. Without replacement, the probability of selecting a second-year student second is $6/11$. The product is $7/12 \times 6/11 = 42/132 = 7/22$.

TSIA2-M5-017 — Correct: C — Correct Answer: 12 percentage points per hour — Explanation: The charge drops by $92 - 56 = 36$ percentage points over 3 hours. The average drop is $36 \div 3 = 12$ percentage points per hour.

TSIA2-M5-018 — Correct: C — Correct Answer: $(x - 4)(x + 4)$ — Explanation: $x^2 - 16$ is a difference of squares: $x^2 - 4^2 = (x - 4)(x + 4)$.

TSIA2-M5-019 — Correct: D — Correct Answer: (3, -3) — Explanation: Moving 7 units right changes the x-coordinate from -4 to 3. Moving 9 units down changes the y-coordinate from 6 to -3.

TSIA2-M5-020 — Correct: B — Correct Answer: The mode is 23 orders. — Explanation: The value 23 appears twice, more often than any other value in the data set.